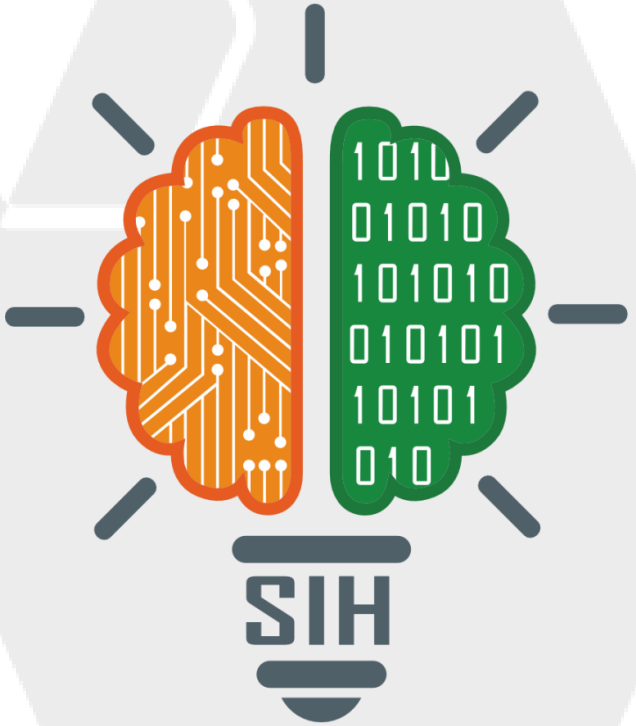
SMART INDIA HACKATHON 2026

TITLE PAGE

- **Problem Statement ID – 26171**
- **Problem Statement Title- On-device Visual Perception for Light-weight Browser Agents**
- **Theme- Smart Automation**
- **PS Category- Software**
- **Team ID-126713**
- **Team Name-BinaryBrainCells**



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IDEA TITLE: VEIL

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A privacy-preserving browser agent: it sees locally, and the cloud model reasons on placeholders only.

Addressing the Problem

- **Screenshot agents:** send everything on screen, personal data included, to a remote model.
- **DOM-only agents:** miss text drawn in a canvas or image.
- **82.5%:** success of attacks extracting sensitive screen info; user profiles inferred ~70% of the time.
- **10:** popular GenAI browser extensions sent page content to their servers.
- **Jan 2025:** Finance Ministry advised staff to avoid ChatGPT and DeepSeek on office devices.

Proposed Solution

- **Perceive (steps 1–3, on device):** reads the page (DOM → IR) and runs OCR locally on only the region that is needed.
- **Protect (steps 4–7, on device):** one sanitizer swaps sensitive values for placeholders; real values stay in an in-memory vault; an egress gate checks every message.
- **Reason (step 8, cloud):** the planner sees placeholders only and proposes one action.
- **Act (steps 9–12, on device):** ten local rules validate it; the browser resolves placeholders, executes and then verifies.

Tech Stack

- **Browser:** Chrome MV3, TypeScript, Side Panel
- **Privacy:** Zod, PII Sanitizer, Vault, Egress Gate
- **On-device AI:** PP-OCRv5, ONNX Runtime Web, WASM, OpenCV.js
- **Planner:** FastAPI, Pydantic, Groq (gpt-oss-20b), Gemini failover
- **Testing:** Vitest, Pytest, Chrome/CDP
- **Monitoring:** SSE, read-only Proof Dashboard

The Unique Gap

Today, a browser agent must choose and see the whole screen and send it to the cloud, or read only the DOM and miss any text drawn as pixels.
VEIL removes this trade-off: it reads pixel-only content locally, and the cloud model reasons on placeholders only.

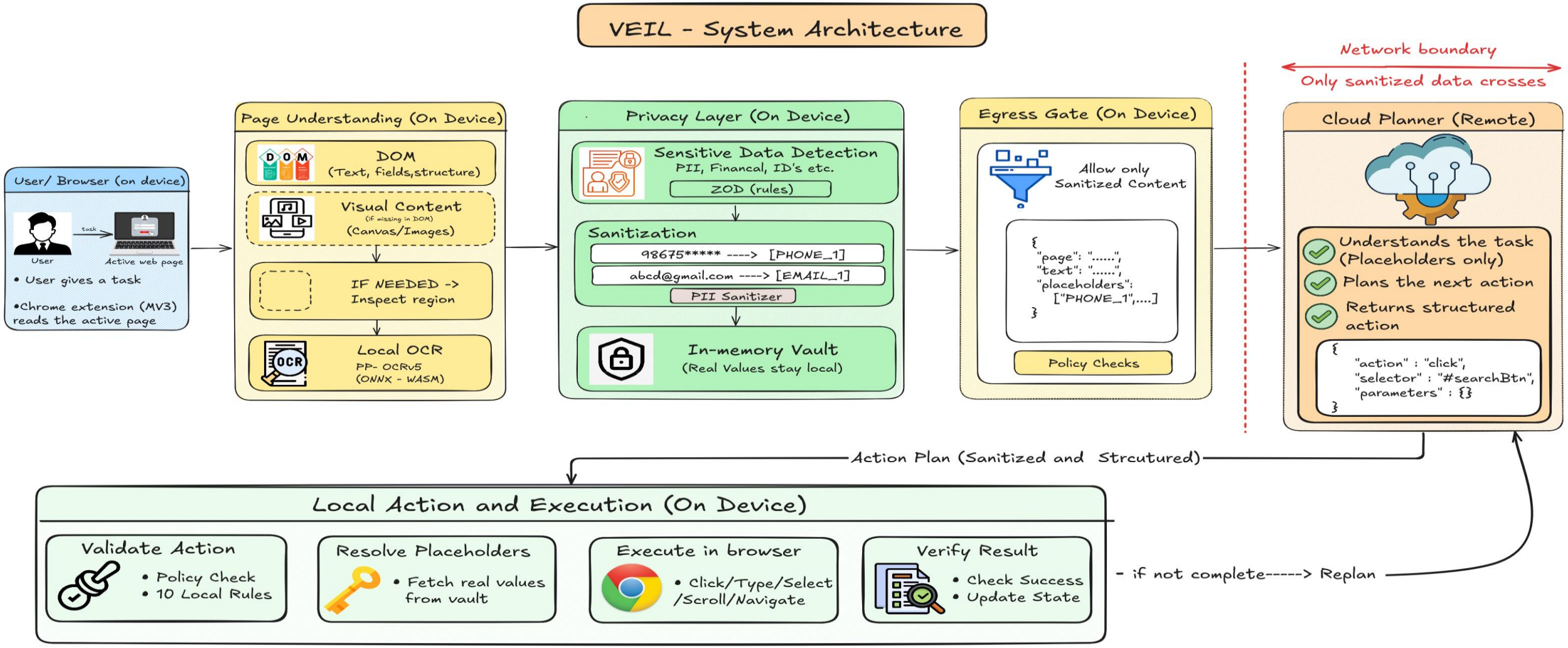
Approach	Reads pixel-only text?	Raw screen data sent to cloud?
Screenshot agents	Yes	Yes, everything on screen
DOM-only agents	No	Page text only
VEIL	**Yes, locally**	**No, placeholders only**

- **Prototype proof (live demo):** a phone number drawn only on a canvas was read locally, sent to the cloud as [PHONE_1], then resolved, typed and verified in the browser.
- **Measured:** 0 of 3 canvas values appeared in the DOM text, raw IR or sanitized view.

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TECHNICAL APPROACH

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Code: github.com/MehulSharmaCode/Veil **Prototype Demo:** https://youtu.be/zFTzaUsmiKA

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FEASIBILITY AND VIABILITY

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327
automated tests pass · TESTED
147 TypeScript · 163 Python · 17 dashboard

0 / 22
canary values leaked · MEASURED
across 180 logged planner payloads

0 / 13
planted values in the IR · MEASURED
none entered the page model

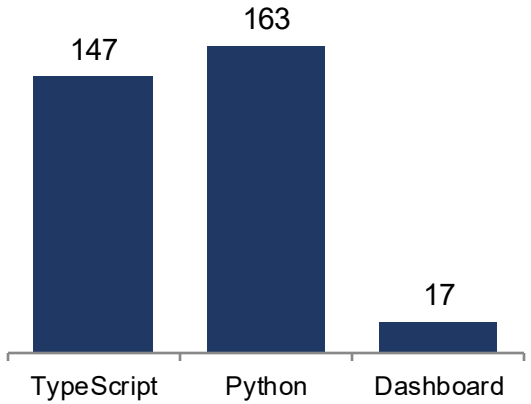
1.0–1.4 s
local OCR per region · MEASURED
5 live runs; no GPU needed

4 / 4
save actions held for the user · LIVE
all denied; none executed

Technical Feasibility

Component	Why it is there
Chrome MV3 + side panel	Runs where users already browse
PP-OCRv5 (4.8 + 16.7 MB)	Reads text that exists only as pixels
ONNX Runtime Web · WASM	Runs on the CPU; no GPU needed
Sanitizer · vault · egress gate	Enforces the privacy boundary
Structured actions + policy	Model proposes; local rules decide

327 Tests by Suite



Risks → Mitigation

Risk	Mitigation
Free-tier rate limits	Retries + Groq to Gemini failover
Heuristic PII detection	Fail-closed gate + canary tests
One controlled page tested	Staged tests: React, then complex sites
OCR-only visual slice	Add face detection and pixel redaction

Operational Feasibility

- Real Chrome extension with side panel
- Local OCR on the device
- Browser execution with verification
- Read-only proof dashboard

India Context

91.9% Chrome share of India's desktop browsers

42% organizations with shadow-AI policies

37% organizations with AI access controls

Sustainability

- **Customer:** institutions with sensitive browser workflows
- **Value:** AI automation without exporting personal data
- **Revenue:** managed licence, on-premise deployment and support

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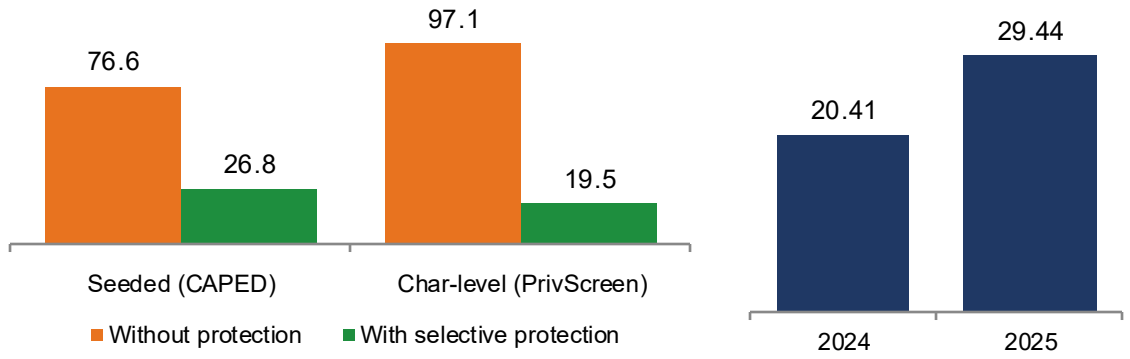
IMPACT AND BENEFITS

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Impact on Government

- **Jan 2025:** Finance Ministry advised staff to avoid ChatGPT and DeepSeek on office devices.
- **IndiaAI's Safe & Trusted AI pillar:** 13 projects selected from 2,000+ proposals (PIB, Oct 2024).
- **VEIL** closes this gap at the browser boundary.
- **Outcome:** By preventing sensitive data exposure at the browser–AI layer, VEIL can help organizations avoid costly breaches, shadow-AI exposure and regulatory penalties.

Research: Leakage (%) and CERT-In Incidents (lakh)



Research-Backed Impact

- **0.766 → 0.268:** task-aware exposure control cut seeded leakage (CAPED, 2026).
- **97.1% → 19.5%:** placeholders cut character-level leakage while accuracy stayed at 89% (Zhao et al., 2026).
- **4.38%:** of image content was flagged as PII, so perception can target only the sensitive part.

Benefits

- **Trust & Governance:** every outbound payload passes one egress gate and is visible on a proof dashboard.
- **Operational & Economic:** local OCR avoids cloud-vision calls; about $0.002–0.005 per 8-step task (estimate).
- **Privacy:** sensitive values stay in a local vault; the design is built around data minimisation.
- **Safety:** the planner proposes, local rules decide, and submit always waits for the user.

Potential Deployment Context

- **72.43 crore DigiLocker · 11.66 crore UMANG** registered users (MeitY, 31 Jul 2026); personal data appears on screen.

Benefits in Numbers

₹25.5 crore average India breach cost (IBM 2026)	**+₹1.79 crore** added by shadow AI (IBM)	**₹250 crore** maximum DPDP penalty (PIB)	**29.44 lakh** CERT-In incidents in 2025

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RESEARCH AND REFERENCES

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PROBLEM EVIDENCE	DESIGN METHODOLOGY & TECHNOLOGY
1. Zhao et al., "Available but Invisible," 2026 Local anonymization protects sensitive GUI data. arxiv.org/abs/2602.10139	**1. Orekondy et al., "Connecting Pixels to Privacy and Utility," CVPR 2018** Studies image redaction privacy-utility tradeoffs. openaccess.thecvf.com (CVPR 2018 paper)
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4. PrivacyLens, NeurIPS 2024 Evaluates privacy risks in agent actions. arxiv.org/abs/2409.00138	**4. PP-OCR, 2020** Lightweight OCR for efficient text extraction. arxiv.org/abs/2009.09941
5. LeakageBench, 2026 Benchmarks PII leakage in document redaction. arxiv.org/abs/2609.02207	**5. "Anatomizing Deep Learning Inference in Web Browsers"** Measures browser-based deep learning performance. arxiv.org/abs/2402.05981